Dynamic Planet – EXAM

2022 UMD Invitational

Team Number: _____

Team/School Name: _____

No abbreviations / PRINT LEGIBLY

Student Names (First & Last): PRINT LEGIBLY

1. _____

2. _____

Total Points Possible (written test): **64**

Total Points Earned:_____

Team Number: _____ School Name: _____

Part I - Multiple Choice (27 pts)

Each question is worth 1 point

1. What do geologists call an area in which all water drains to the same location?
    a. Drainage region
    b. Drainage basin
    c. Drainage channel
    d. Drainage delta

2. What channel type is likely to form in a slow-moving, shallow part of a river with high sediment deposition?
    a. Meandering channel
    b. Straight channel
    c. Braided channel
    d. Anastomosing channel

3. A parallel drainage pattern would most likely form:
    a. Around an elevated point
    b. On the side of a slope
    c. Above a layer of homogenous bedrock
    d. In a region that has undergone excessive faulting

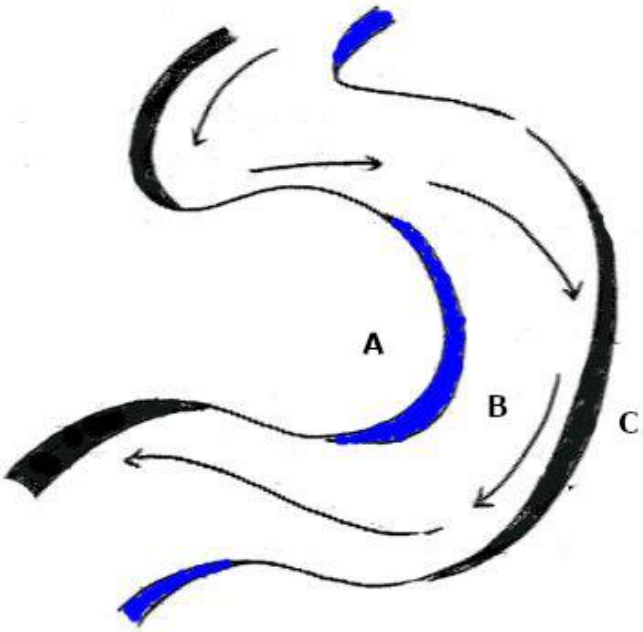
4. Which factor has the most influence on stream competence?
    a. Discharge
    b. Depth
    c. Velocity
    d. Gradient

5. What is the term for the maximum total load that a stream can move?
    a. Stream competence
    b. Stream capability
    c. Stream cohesion
    d. Stream capacity

6. Which of the following is the largest clast type?
    a. Gravel
    b. Silt
    c. Clay
    d. Sand

7. Which type of sediment load is carried along with water in the stream?
    a. Bedload
    b. Dissolved load
    c. Mixed load
    d. Suspended load

8. Which location in the below image will experience the most erosion?



    a. A
    b. B
    c. C
    d. A and C

9. Streams that only occasionally have water flowing in them are called:
    a. Perennial streams
    b. Ephemeral streams
    c. Annual streams
    d. Temporary streams

10. Which of the following is NOT a cause of rapids?
    a. Increased discharge
    b. Increased gradient
    c. Abrupt channel narrowing
    d. Stream flows over large clasts

11. Which conditions are necessary for a canyon to form?
    a. Slow downcutting and soft, erodible rock.
    b. Rapid downcutting and soft, erodible rock.
    c. Rapid downcutting and hard, erosion resistant rock.
    d. Slow downcutting and hard, erosion resistant rock.

12. The Zone of Aeration is the part of the ground where gaps between the soil are filled by:
    a. Air only
    b. Water only
    c. Both air and water
    d. Neither air nor water

13. What is the term for the rate at which water flows through soil?
    a. Porosity
    b. Conductivity
    c. Capacity
    d. Permeability

14. An aquifer bounded on top by the water table and below by a confining bed is referred to as a(n):
    a. Confined aquifer
    b. Unconfined aquifer
    c. Semi-confined aquifer
    d. Confining aquifer

15. Karst is an area of land where the bedrock is made of:
    a. Sandstone
    b. Dolomite
    c. Limestone
    d. Both b and c are possible

16. What process forms kettle lakes?
    a. Faulting
    b. Volcanic activity
    c. Rifting
    d. Glaciation

17. Lakes formed by tectonic activity form in what kind of basin?
    a. Graben
    b. Rift
    c. Forearc basin
    d. River basin

18. Which of the following is *not* part of the three-layer structure of a lake?
    a. Epilimnion
    b. Mesolimnion
    c. Metalimnion
    d. Hypolimnion

19. What is the main source of energy in a lake?
    a. Geothermal heat
    b. Photosynthesis
    c. Wind
    d. Solar radiation

20. What feature of wetlands is used to classify them as a marsh, swamp, bog, or fen?
    a. Salinity
    b. Vegetation
    c. Water depth
    d. Climate

21. Which of the following is *not* an effect that overgrazing by animals has on wetlands?
    a. Removal of vegetation
    b. Stream bank destabilization
    c. Soil compaction
    d. Lowering of the water table

22. Which of the following is an effect dams have downstream?
    a. Lower nutrient content
    b. Greater erosion
    c. Larger clast size
    d. All of the above

23. The process through which leaves transport water to the atmosphere is called:
    a. Evaporation
    b. Transpiration
    c. Sublimation
    d. Infiltration

24. The leading cause of water pollution in the United States is:
    a. Point pollution
    b. Nonpoint pollution
    c. Transboundary pollution
    d. None of the above

25. Over ____ of wastewater in the world flows back into the environment without being treated.
    a. 80%
    b. 85%
    c. 90%
    d. 95%

26. Permafrost is defined as ground that remains frozen for at least:
    a. One year
    b. Two years
    c. Five years
    d. Ten years

27. Infiltration is defined as:
    a. Pollutants entering a water supply
    b. Salt water contaminating an artesian well
    c. Water on the surface entering the soil
    d. A stream overflowing onto the floodplain

Part II - Short Answer (37 pts)

Point values specified for each question

1. A stream runs between two points that are 10 km apart. The stream channel covers a total distance of 17 km, and the elevation difference between the two points is 500 m.
    a. What is the sinuosity of the stream? (1)

    b. What is the channel type of the stream? (1)

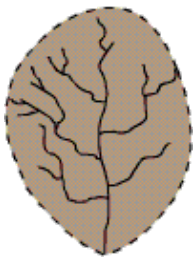
2. How do bedload particles move? What is this process called? (3)

3. Why do rivers that frequently flood form natural levees? (2)

4. How do waterfalls form? Why do waterfalls tend to retreat over time? (4)

5. What is an alluvial fan? How do they form? (2)

6. Label each of the following drainage patterns (2: 0.5 pts each)


a.


b.


c.


d.

a. _____

b. _____

c. _____

d. _____

7. Geologists want to study two underground points that are 10 km apart. Each point is 1000m above sea level. They drill wells at each point. At point A, the groundwater rises to 1434m above sea level. At point B, the water rises to 1796m.
   a. What direction is the groundwater flowing? (1)

   b. What is the hydraulic head gradient from point A to point B? (2)

Answer: _____

c. The geologists find that the rate of flow between the two points through an area of 100 m^2 is $3*10^{-6}$ m^3/s. Find the hydraulic conductivity, K (2).

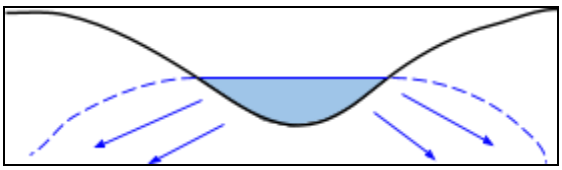
Answer: _____

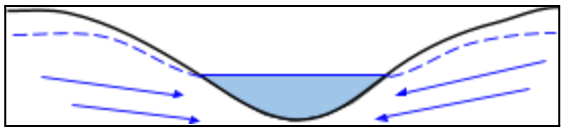
d. Based on this information, which of the following is the most likely type of rock found between the two points? (1)
   i. Limestone
   ii. Siltstone
   iii. Clay
   iv. Shale

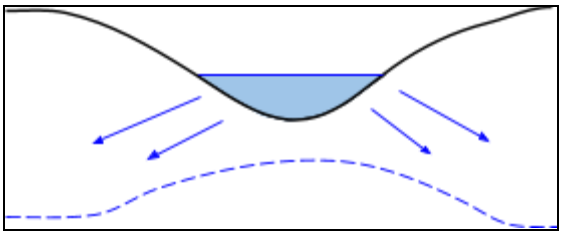
8. Describe the process of stream piracy. (2)

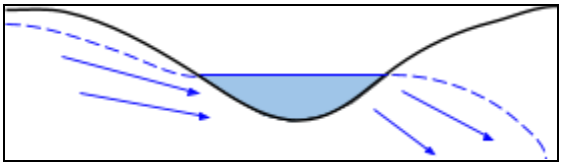
9. Describe three ways in which dams harm fish populations. (3)

10. Label each type of wetland based on its interaction with groundwater. (4)

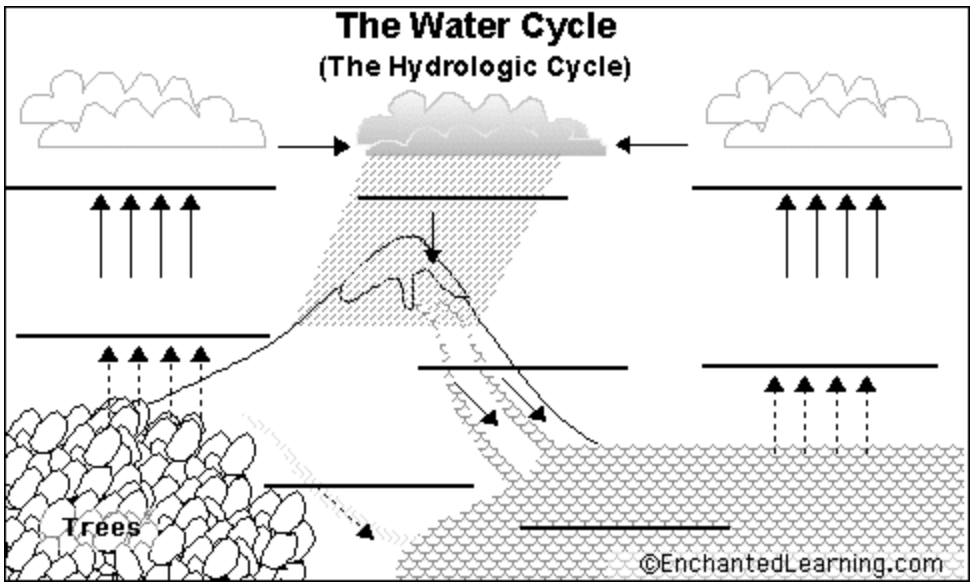
a.  _____

b.  _____

c.  _____

d.  _____

11. Label the different steps of the hydrologic cycle in the diagram below. (4: 0.5 pts each)



12. While flying over Russia in the Arctic Circle, you spot the landform seen in this picture. What is it called? How does it form? (3)

